

Practice Test II (Version 1)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
 II) Show all work.
 III) Draw a rectangle around your final answer for each problem.
 IV) You may use a calculator except for graphing.

① Graph the following equation. Write the coordinates of the foci.

$$\frac{(x+1)^2}{9} + \frac{(y-1)^2}{4} = 1$$

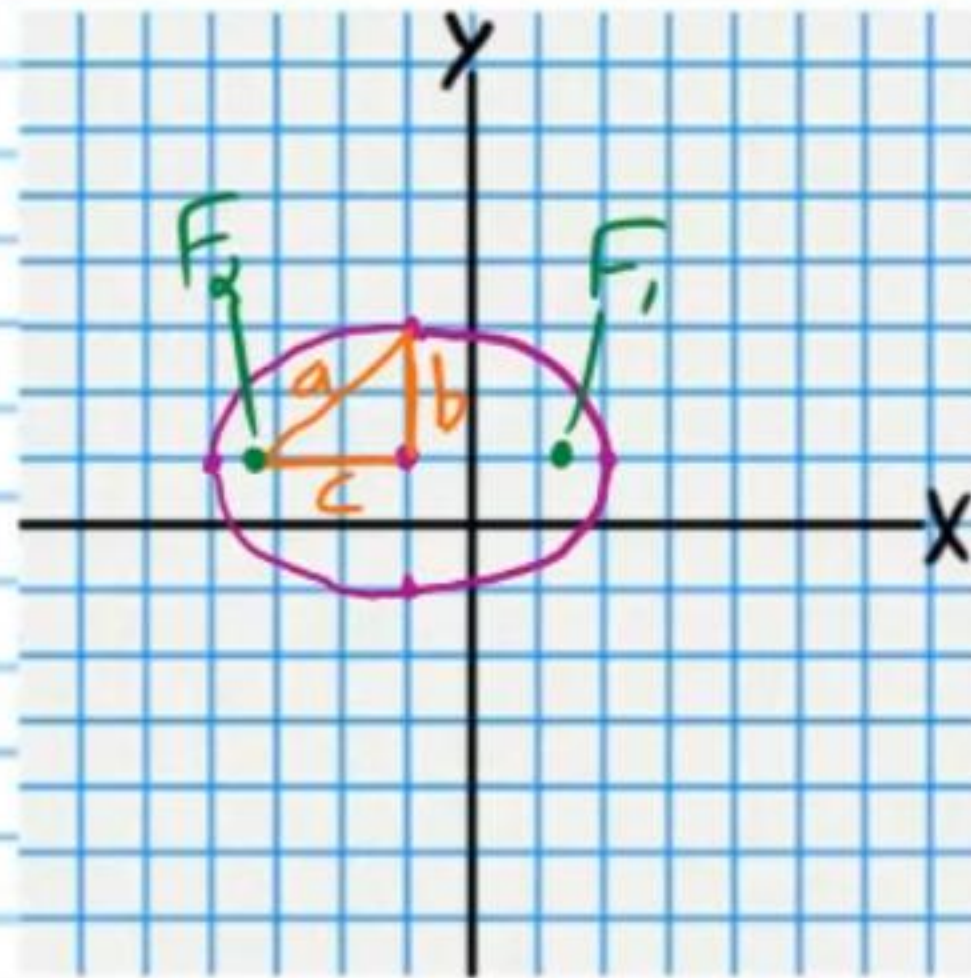
$$a^2 = b^2 + c^2$$

$$3^2 = 2^2 + c^2$$

$$c = \sqrt{5}$$

$$F_2(-1-\sqrt{5}, 1)$$

$$F_1(-1+\sqrt{5}, 1)$$



② Find the sums below.

$$i) \sum_{n=3}^{\infty} \left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^{3-1} + \left(\frac{1}{2}\right)^{4-1} + \left(\frac{1}{2}\right)^{5-1} + \dots$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$S_{\infty} = \frac{a_1}{1-r}$$

$$S_{\infty} = \frac{\frac{1}{4}}{1-\frac{1}{2}}$$

$$S_{\infty} = \left[\frac{1}{2}\right]$$

$$ii) \sum_{n=1}^4 3^n = 3^1 + 3^2 + 3^3 + 3^4$$

$$= 3 + 9 + 27 + 81$$

$$= \boxed{120}$$

③ Graph the following equation. Write the coordinates of the foci, draw the asymptotes as dotted lines, and write their equations.

$$1 = \frac{y^2}{25} - (x-2)^2$$

$$1 = \frac{(y-0)^2}{25} - (x-2)^2$$

Find the foci

$$a^2 + b^2 = c^2$$

$$5^2 + 1^2 = c^2$$

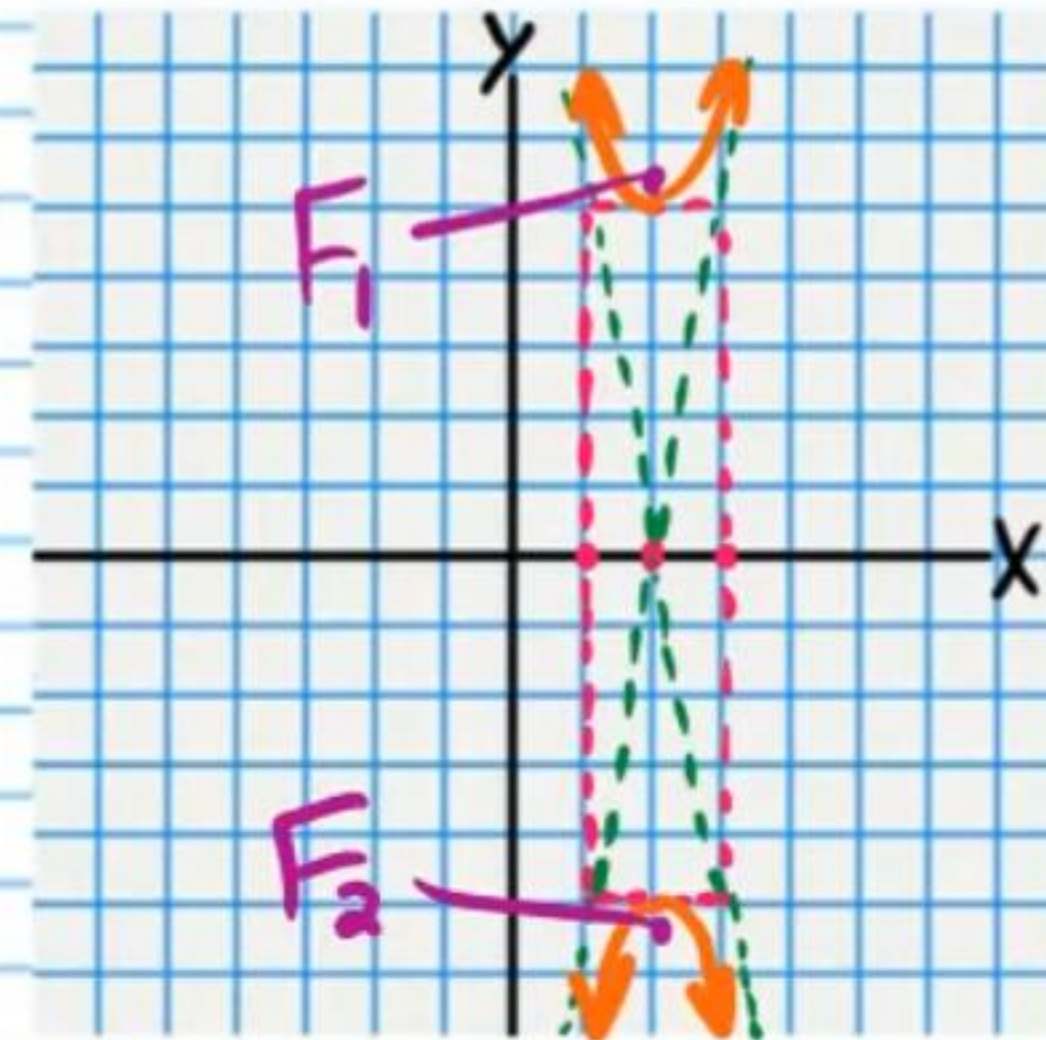
$$c = \sqrt{26}$$

$$F_1(2, 0 + \sqrt{26})$$

$$F_2(2, 0 - \sqrt{26})$$

$$F_1(2, \sqrt{26})$$

$$F_2(2, -\sqrt{26})$$



Asymptotes

$$y - y_p = m(x - x_p)$$

$$P(2, 0)$$

$$y - 0 = 5(x - 2)$$

$$\boxed{y = 5x - 10}$$

$$y - 0 = -5(x - 2)$$

$$\boxed{y = -5x + 10}$$

④ Write an equation of the graph below. "P" is a point on the parabola and "V" is the vertex.

$$V(-3, -1), P(-6, 1)$$

$$y - k = a(x - h)^2$$

$$1 - (-1) = a(-6 - (-3))^2$$

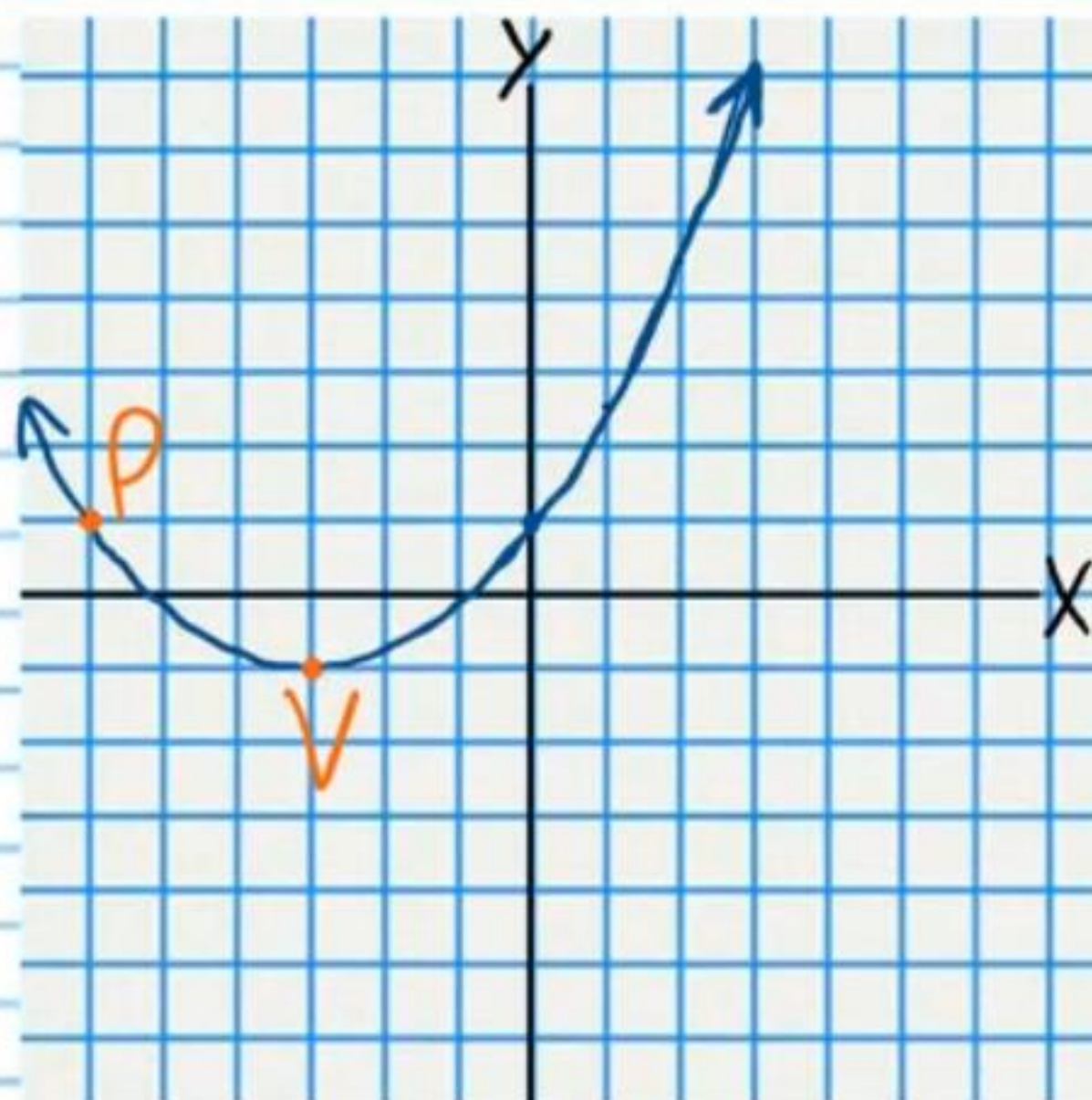
$$1 + 1 = a(-6 + 3)^2$$

$$2 = a(-3)^2$$

$$2 = 9a$$

$$a = \frac{2}{9}$$

$$y + 1 = \frac{2}{9}(x + 3)^2$$



⑤ Find an equation of the hyperbola with the following characteristics and graph the equation.

Foci at $(0, -2)$ and $(4, -2)$, vertex at $(3, -2)$.

$$\text{Center} = (2, -2) \quad \frac{(x-2)^2}{1^2} - \frac{(y+2)^2}{(\sqrt{3})^2} = 1$$

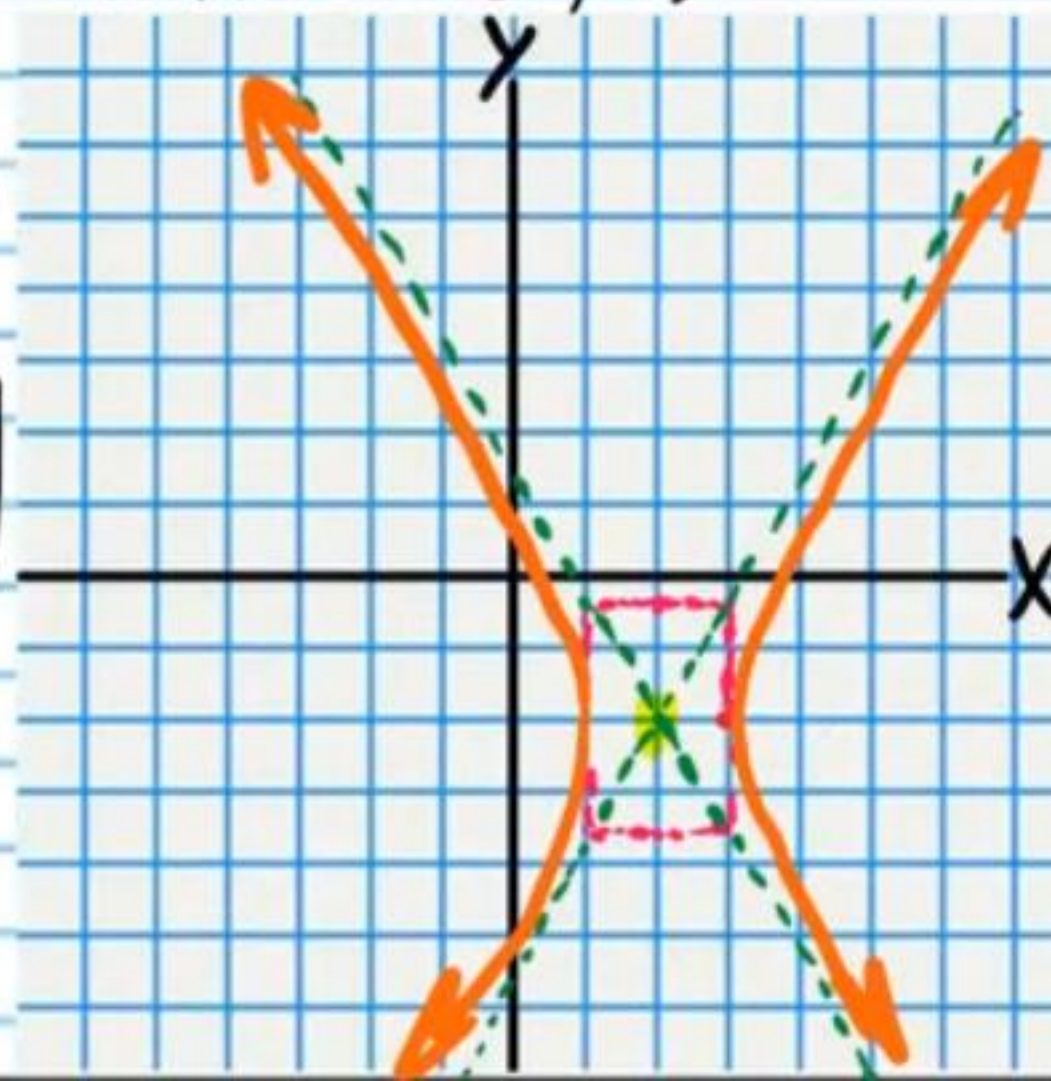
$$a = 1, c = 2$$

$$a^2 + b^2 = c^2$$

$$1^2 + b^2 = 2^2$$

$$b = \sqrt{3}$$

$$\frac{(x-2)^2}{1} - \frac{(y+2)^2}{3} = 1$$



⑥ Graph the following equation. *You must first convert to a recognizable form.*

$$36x^2 + 5y^2 - 72x - 10y - 139 = 0$$

$$36x^2 - 72x + 5y^2 - 10y = 139$$

$$36(x^2 - 2x) + 5(y^2 - 2y) = 139$$

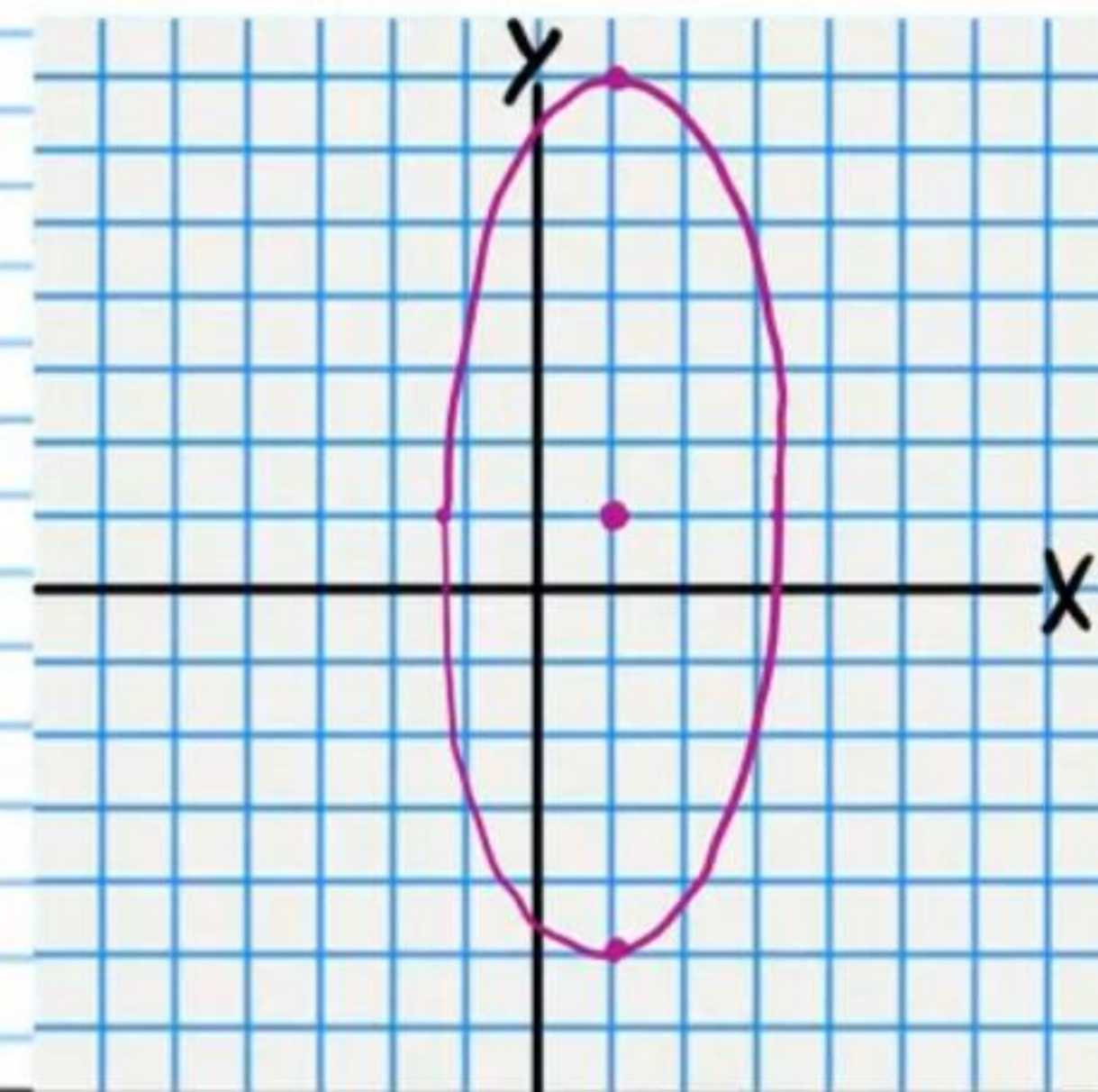
$$36(x^2 - 2x + 1) + 5(y^2 - 2y + 1) = 139 + 36 + 5$$

$$36(x-1)(x-1) + 5(y-1)(y-1) = 180$$

$$36(x-1)^2 + 5(y-1)^2 = 180$$

$$\frac{36(x-1)^2}{180} + \frac{5(y-1)^2}{180} = \frac{180}{180}$$

$$\frac{(x-1)^2}{5} + \frac{(y-1)^2}{36} = 1$$



⑦ Graph the following equation. *You must first convert to a recognizable form.*

$$x^2 + y^2 + 4x - 6y - 3 = 0$$

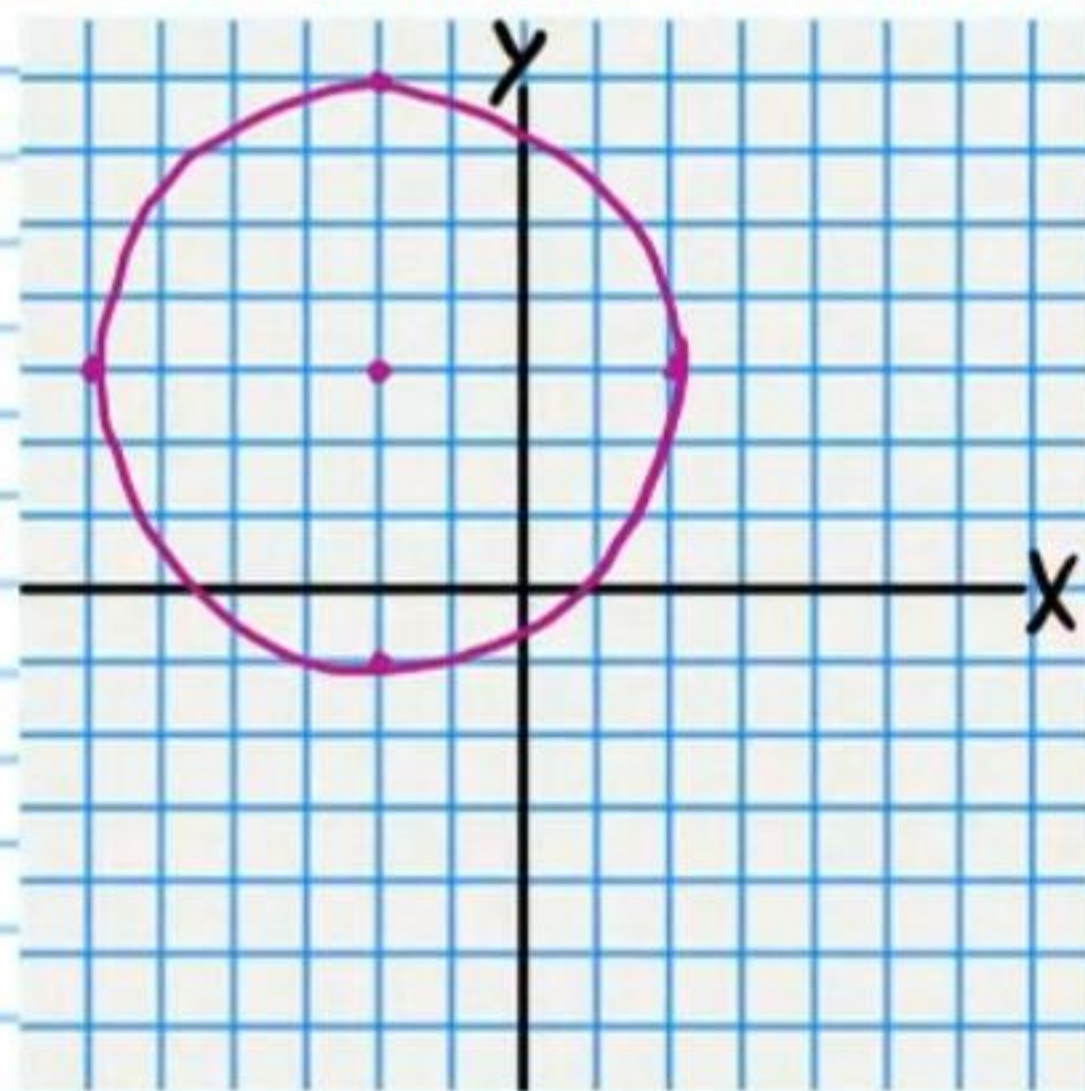
$$x^2 + 4x + y^2 - 6y = 3$$

$$x^2 + 4x + 4 + y^2 - 6y + 9 = 3 + 4 + 9$$

$$(x+2)(x+2) + (y-3)(y-3) = 16$$

$$(x+2)^2 + (y-3)^2 = 16$$

$$(x-h)^2 + (y-k)^2 = r^2$$



⑧ Find the 75th term in the arithmetic sequence below.

$$3, -8, -19, \dots$$

$$a_n = a_1 + D(n-1)$$

$$a_{75} = 3 - 11(75-1)$$

$$a_{75} = 3 - 11(74)$$

$$a_{75} = 3 - 814$$

$$\boxed{a_{75} = -811}$$

⑨ Use the Binomial Theorem to expand the following expression.

$$(m-2)^5 = (m+(-2))^5$$

$$\binom{5}{0}m^5(-2)^0 + \binom{5}{1}m^4(-2)^1 + \binom{5}{2}m^3(-2)^2 + \binom{5}{3}m^2(-2)^3 + \binom{5}{4}m^1(-2)^4 + \binom{5}{5}m^0(-2)^5$$

$$(1)m^5(1) + 5m^4(-2) + 10m^3(4) + 10m^2(-8) + 5m(16) + (1)(1)(-32)$$

$$\boxed{m^5 - 10m^4 + 40m^3 - 80m^2 + 80m - 32}$$

⑩ Find the sum.

$$\sum_{n=0}^{93} 2n = 2(0) + 2(1) + 2(2) + 2(3) = 0 + 2 + 4 + 6 + \dots$$

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{94} = \frac{94}{2}(2(0) + 2(94-1))$$

$$= 47(0 + 2(93))$$

$$= 47(186)$$

$$= \boxed{8,742}$$

⑪ There are 44 students in a class. 19 of them like pizza. 30 of them like tacos. 12 of them like both.

i) How many students like pizza but not tacos?

$$19 - 12 = \boxed{7 \text{ students}}$$

ii) How many students like tacos but not pizza?

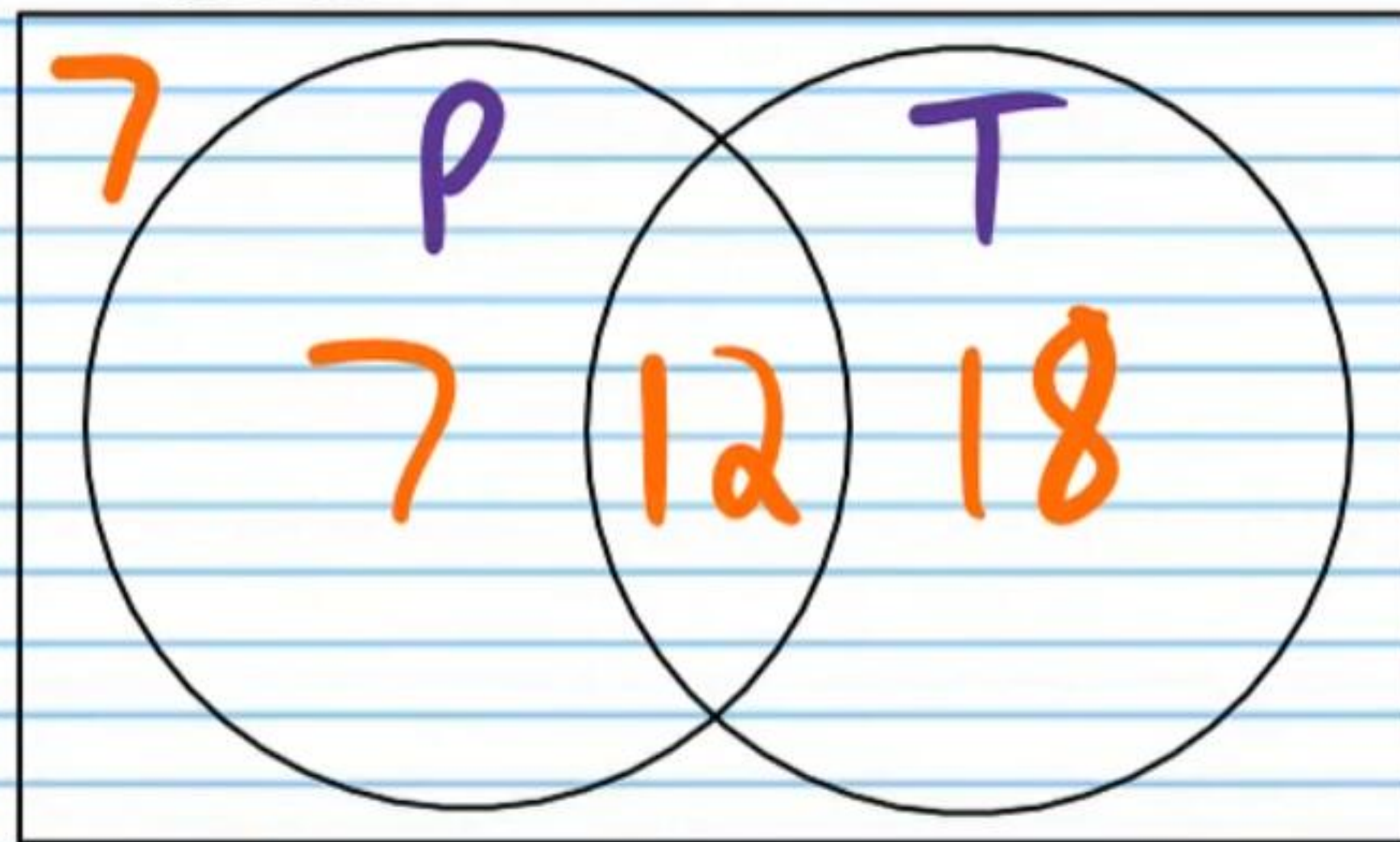
$$30 - 12 = \boxed{18 \text{ students}}$$

iii) How many students like **either** pizza **or** tacos?

$$7 + 12 + 18 = \boxed{37 \text{ students}}$$

iv) How many students like **neither** pizza **nor** tacos?

$$44 - 37 = \boxed{7 \text{ students}}$$



v) What is the probability that a randomly selected student will like pizza?

$$\boxed{\frac{19}{44}}$$

vi) What is the probability that a randomly selected student will like **neither** pizza **nor** tacos?

$$\boxed{\frac{7}{44}}$$

vii) What is the probability that a randomly selected student will like **both** pizza **and** tacos?

$$\frac{12}{44} = \boxed{\frac{3}{11}}$$

viii) What is the probability that a randomly selected student will like **either** pizza **or** tacos?

$$\boxed{\frac{37}{44}}$$

⑫ i) Find the sum of the first 37 terms in the geometric sequence below. You may leave your answer in exponential form if parts are very large or very small.

$$\frac{1}{2}, -\frac{1}{3}, \frac{2}{9}, \dots$$

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{37} = \left(\frac{1}{2}\right) \frac{1 - \left(-\frac{2}{3}\right)^{37}}{1 - \left(-\frac{2}{3}\right)}$$

$$S_{37} = \left(\frac{1}{2}\right) \frac{1 + \left(\frac{2}{3}\right)^{37}}{1 + \frac{2}{3}}$$

$$S_{37} = \left(\frac{1}{2}\right) \frac{1 + \left(\frac{2}{3}\right)^{37}}{\frac{5}{3}}$$

$$S_{37} = \frac{1 + \left(\frac{2}{3}\right)^{37}}{\frac{10}{3}}$$

$$S_{37} = \left(1 + \left(\frac{2}{3}\right)^{37}\right) \div \frac{10}{3}$$

$$S_{37} = \left(1 + \left(\frac{2}{3}\right)^{37}\right) \cdot \frac{3}{10}$$

$$S_{37} = \frac{3\left(1 + \left(\frac{2}{3}\right)^{37}\right)}{10}$$

ii) Find the 41st term in the geometric sequence below. You may leave your answer in exponential form if it is very large or very small.

$$-\frac{1}{2}, -2, -8, \dots$$

$$a_n = a_1 r^{n-1}$$

$$a_{41} = -\frac{1}{2}(4)^{41-1}$$

$$a_{41} = -\frac{1}{2}(4)^{40}$$

⑬ Graph the following equation. Write the coordinates of the focus and the equation of the directrix.

$$x-2 = -2(y-4)^2$$

Focus

$$p = \frac{1}{4a}$$

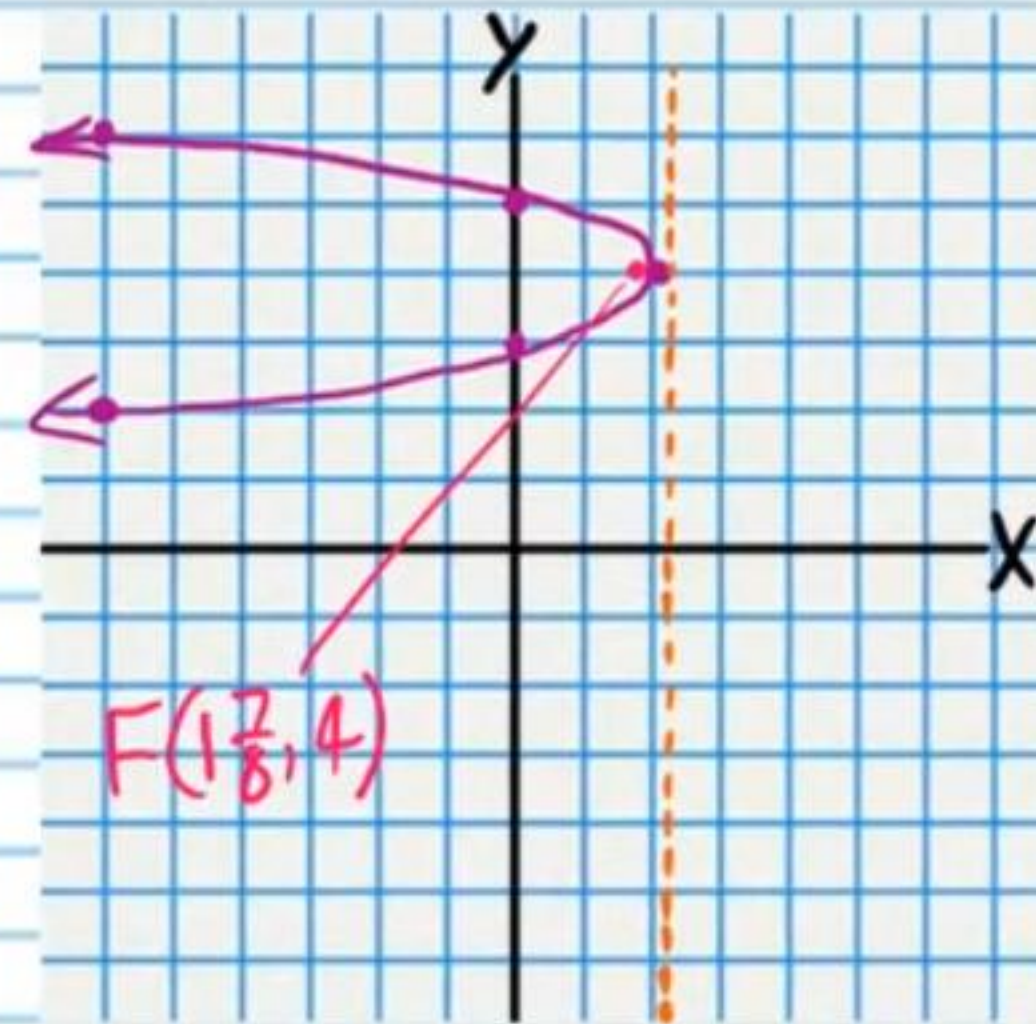
$$p = \frac{1}{4(-2)}$$

$$p = -\frac{1}{8}$$

$$F\left(1\frac{7}{8}, 4\right)$$

Directrix

$$x = 2\frac{1}{8}$$



⑭ i) There are 24 letters in the Korean alphabet. How many outcome variations are there for a four letter code if no letter can be used more than once?

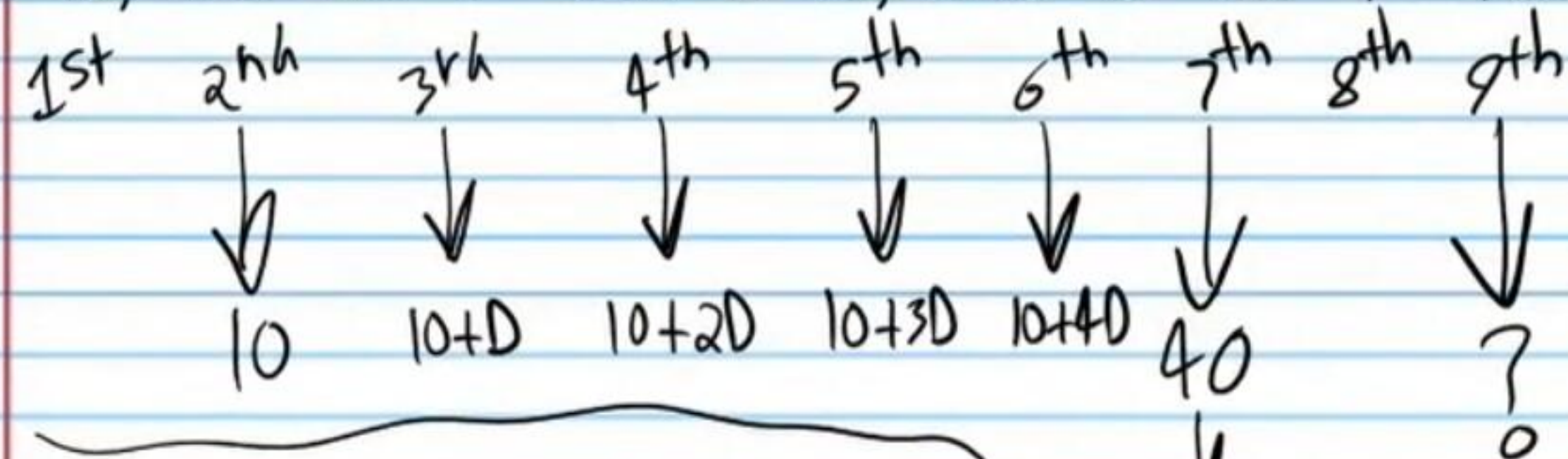
$$P(24, 4) = {}_nP_r = 24 \times 23 \times 22 \times 21 = 255,024 \text{ outcome variations}$$

ii) Write an equation of the n^{th} term of the sequence below in recursive form. Also, show the value of the first term.

$$1, \frac{5}{3}, \frac{25}{9}, \dots$$

$$\{a_1 = 1, a_n = \frac{5}{3}a_{n-1}\}$$

⑮ If the 2nd term of an arithmetic sequence is 10, and the 7th term is 40, what is the 9th term?



$$\begin{array}{ccc}
 7^{\text{th}} & 8^{\text{th}} & 9^{\text{th}} \\
 \downarrow & \downarrow & \downarrow \\
 40 & 40+D & 40+D+D = \boxed{52}
 \end{array}$$

$$\begin{array}{c}
 10+5D \\
 \downarrow \\
 40 = 10+5D \\
 D = 6
 \end{array}$$

For problem #'s 16-17, determine if each sequence is arithmetic, geometric, or neither. Then find an expression for the n^{th} term. Assume that the first term is produced when $n=1$.

⑮ i) 0.8, 0.2, 0.05, ...

A) Geometric

B) $a_n = a_1 r^{n-1}$
 $a_n = 0.8 \left(\frac{1}{4}\right)^{n-1}$

$$\boxed{\sum 0.8 \left(\frac{1}{4}\right)^{n-1}}$$

ii) $\cos 3, \cos 4, \cos 5, \dots$

A) Neither

B) $\boxed{\sum \cos(n+2)}$

⑮ i) $\log_2 18, \log_2 19, \log_2 20, \dots$

A) Neither

B) $\boxed{\sum \log_2(n+17)}$

ii) -11, 10, 31, ...

A) Arithmetic

B) $a_n = a_1 + D(n-1)$

$$a_n = -11 + 21(n-1)$$

$$a_n = -11 + 21n - 21$$

$$a_n = -32 + 21n$$

$$\boxed{\sum -32 + 21n}$$

⑮ Find the coefficient of the x^6 term in the expansion of $(2x+3)^9$.

$$\binom{9}{0}(2x)^9(3^0) + \binom{9}{1}(2x)^8(3^1) + \binom{9}{2}(2x)^7(3^2) + \binom{9}{3}(2x)^6(3^3) + \dots$$

$$\begin{array}{c}
 \downarrow \\
 (84)(64x^6)(27)
 \end{array}$$

$$\begin{array}{c}
 \downarrow \\
 145,152x^6
 \end{array}$$

$$\begin{array}{c}
 \downarrow \\
 \boxed{145,152}
 \end{array}$$

①9 Johnny has a rock collection. 4 of his rocks are basalt, 5 of them are limestone, and 6 of them are quartz. He chooses 3 of his rocks **at random** to show to his class at school.

i) What is the probability that all 3 are basalt?

$$P(BBB) = \frac{{}^4C_3}{{}^{15}C_3} = \boxed{\frac{4}{455}}$$

ii) What is the probability that 2 rocks will be quartz and 1 will be basalt?

$$P(2 \text{ quartz } 1 \text{ basalt}) = \frac{{}^6C_2 \cdot {}^4C_1}{{}^{15}C_3} = \frac{15 \cdot 4}{455} = \boxed{\frac{12}{91}}$$

iii) What is the probability that 2 will be basalt and 1 will be limestone?

$$P(2 \text{ basalt } 1 \text{ limestone}) = \frac{{}^4C_2 \cdot {}^5C_1}{{}^{15}C_3} = \frac{6 \cdot 5}{455} = \boxed{\frac{6}{91}}$$

iv) What is the probability that 1 rock will be basalt, 1 will be quartz, and 1 will be limestone?

$$P(1 \text{ basalt } 1 \text{ quartz } 1 \text{ limestone}) = \frac{{}^4C_1 \cdot {}^5C_1 \cdot {}^6C_1}{{}^{15}C_3} = \frac{4 \cdot 5 \cdot 6}{455} = \boxed{\frac{24}{91}}$$

②0 A fish is chosen at random from a tide pool. The probability that the fish will be a minnow is 18%. The probability that the fish will be a mackerel is 8%. What is the probability that the fish will be either a minnow or a mackerel?

$$P(\text{minnow} \cup \text{mackerel}) = P(\text{minnow}) + P(\text{mackerel}) = 0.18 + 0.08 = \boxed{0.26}$$

②1 The probability that a football team will win a particular game is 73%. What is the probability that the team will lose?

$$P(W) = 1 - P(\text{not } W)$$

$$0.73 = 1 - P(\text{not } W)$$

$$-0.27 = -P(\text{not } W)$$

$$P(\text{not } W) = \boxed{0.27}$$

②2 Five children are asked to state their favorite character in a comic book. There are 7 characters in the comic book. How many outcome variations are possible?

$$7 \times 7 \times 7 \times 7 \times 7 = 7^5 = \boxed{16,807 \text{ outcome variations}}$$

②③ Sarah has 108 coins. 22 of the coins are made of silver and 56 of them are quarters. 67 of the coins are either quarters or made of silver.

i) How many coins are neither quarters nor made of silver?

$$108 - 67 = \boxed{41 \text{ coins}}$$

ii) How many coins are silver quarters?

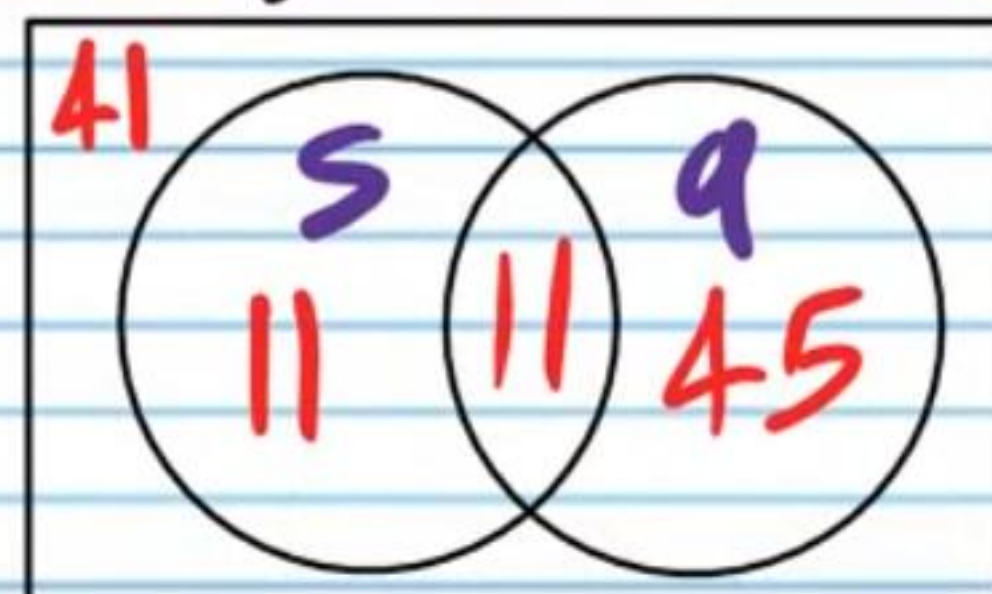
$$n(S \cup Q) = n(S) + n(Q) - n(S \cap Q)$$

$$67 = 22 + 56 - n(S \cap Q)$$

$$67 = 78 - n(S \cap Q)$$

$$-11 = -n(S \cap Q)$$

$$n(S \cap Q) = \boxed{11 \text{ coins}}$$



iii) How many coins are quarters that are not made of silver?

$$56 - 11 = \boxed{45 \text{ coins}}$$

iv) How many coins are made of silver but are not quarters?

$$22 - 11 = \boxed{11 \text{ coins}}$$

②④ A peace organization has 7 doctors, 10 nurses, 3 scientists, and 6 volunteers (with no special skills). The organization will choose a team of 4 doctors, 6 nurses, 2 scientists, and 3 volunteers to help some sick people in an impoverished country. How many outcome variations are possible for the team that the organization will send?

1st choice 2nd choice 3rd choice 4th choice

$${}^7C_4 \quad \times \quad {}^{10}C_6 \quad \times \quad {}^3C_2 \quad \times \quad {}^6C_3$$

$$35 \quad \times \quad 210 \quad \times \quad 3 \quad \times \quad 20$$

$$\boxed{441,000 \text{ outcomes variations}}$$

②⑤ There are 101 supermarkets in Oakdale County. 68 have a coffee shop inside them. 55 have a deli inside them. 30 have a bakery and a deli. 34 have a coffee shop and a deli. 26 have a coffee shop and a bakery. 12 of them have all three. 8 of them have none of these features. How many supermarkets have a bakery but no coffee shop or deli?

$$101 - 8 - 20 - 22 - 12 - 14 - 3 - 18 = \boxed{4 \text{ supermarkets}}$$

